04 TOC 200Q

Topic 4 Theory of Computation - 200 MCQs

Syllabus: Finite Automata, PDA, NFA/DPDA, non-determinism, languages accepted, Grammars, Non-computability, examples of non-computable problems. Includes DFA/NFA conversions, pumping lemma, Chomsky hierarchy, Turing machines, decidability/halting. ~200 Qs.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

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Q1. A DFA is formally a 5-tuple (Q, ?, ?, q0, F) where ? is?

A. Q × ? ?

Q2. An NFA transition function maps to?

A. Single s

Q3. Every NFA can be converted to an equivalent DFA with at most?

A. n! state

Q4. ?-closure of a state includes?

A. Only fir

Q5. Language accepted by DFA is called?

A. Contex

Q6. Pumping lemma for regular languages states: for long enough string xyz with |y|>0, |xy|?p, then?

A. p must

Q7. Which language is NOT regular?

A. (01)*

Q8. Myhill-Nerode theorem characterizes regularity via?

A. Finite i

Q9. Minimal DFA is unique up to isomorphism and found by?

A. Remov

Q10. Regular expressions, NFA, DFA are?

A. Only R

Q11. Context-free grammars are recognized by?

A. Finite a

Q12. A DPDA is strictly weaker than?

A. NFA

Q13. Example of deterministic CFL but ambiguous grammar?

A. a^n b^n

Q14. Chomsky Normal Form productions are of type?

A. A?? on

Q15. Greibach Normal Form productions start with?

A. Termin

Q16. Chomsky hierarchy from most restricted: Regular ? CFL ? CSL ? ?

A. Finite o

Q17. Type-0 grammars correspond to?

A. PDA

Q18. Type-1 (context-sensitive) correspond to?

A. Linear

Q19. Pumping lemma for CFL uses?

A. No bou

Q20. Which language is CFL but not regular?

A. {ww | w

Q21. Ambiguous grammar means?

A. Infinite

Q22. Inherent ambiguity example?

A. (a+b)*

Q23. CYK algorithm decides membership for CFG in CNF in?

A. O(n)

Q24. Pushdown automaton uses?

A. Stack (

Q25. Turing machine adds over PDA?

A. Only st

Q26. Church-Turing thesis states?

A. P=NP

Q27. Halting problem is?

A. Regula

Q28. An example non-computable / undecidable problem?

A. Sorting

Q29. Rice's theorem says?

A. TMs alw

Q30. Decidable problems for DFA include?

A. Halting

Q31. Which CFL problem is undecidable?

A. Membe

Q32. Post Correspondence Problem (PCP) is?

A. Finite

Q33. Linear bounded automata decide?

A. Regula

Q34. A universal Turing machine?

A. Cannot

Q35. Recursively enumerable vs recursive?

A. Recurs

Q36. Complement of halting language is?

A. CFL

Q37. If L and its complement are both RE then L is?

A. Recurs

Q38. Kleene closure L* is regular if L is regular: proof uses?

A. ?-NFA

Q39. Arden's theorem solves?

A. Equatio

Q40. Mealy vs Moore machines differ in?

A. Output

Q41. The number of states in minimal DFA for language (0+1)*0(0+1){2} (k fixed suffix length) is at least?

A. 8

Q42. Consider the standard result: An NFA transition function maps to?

A. Stack s

Q43. L = {a^n b^n c^n} is?

A. Regula

Q44. If NFA has 3 states, worst-case DFA via subset construction has?

A. 7 states

Q45. Consider the standard result: Language accepted by DFA is called?

A. Recurs

Q46. Intersection of CFL with regular is?

A. Regula

Q47. Pumping length p for DFA with 3 states can be taken as?

A. 4

Q48. Consider the standard result: Myhill-Nerode theorem characterizes regularity via?

A. Stack d

Q49. L = {a^p | p prime} is?

A. Star-fre

Q50. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Consta

Q51. Consider the standard result: Context-free grammars are recognized by?

A. Turing

Q52. Empty-string membership for RE language is?

A. Always

Q53. For alphabet size 3, number of strings of length n is?

A. 3^n

Q54. Consider the standard result: Chomsky Normal Form productions are of type?

A. A?a*

Q55. Complement of CFL is?

A. Not nec

Q56. The number of states in minimal DFA for language (0+1)*0(0+1){2} (k fixed suffix length) is at least?

A. 8

Q57. Consider the standard result: Type-0 grammars correspond to?

A. Turing

Q58. L = {ww^R | w?{a,b}*} is?

A. Determ

Q59. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5^2 sta

Q60. Consider the standard result: Which language is CFL but not regular?

A. {a^n b^

Q61. Equivalence of two DFAs is decidable by?

A. Minimiz

Q62. Pumping length p for DFA with 3 states can be taken as?

A. 3

Q63. Consider the standard result: CYK algorithm decides membership for CFG in CNF in?

A. O(2^n)

Q64. Every finite language is?

A. Not reg

Q65. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Expone

Q66. Consider the standard result: Church-Turing thesis states?

A. P=NP

Q67. L = {a^n b^n c^n} is?

A. CFL

Q68. For alphabet size 3, number of strings of length n is?

A. 3+n

Q69. Consider the standard result: Rice's theorem says?

A. P=NP

Q70. Intersection of CFL with regular is?

A. Undeci

Q71. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 17

Q72. Consider the standard result: Post Correspondence Problem (PCP) is?

A. Decida

Q73. L = {a^p | p prime} is?

A. Star-fre

Q74. If NFA has 4 states, worst-case DFA via subset construction has?

A. 2^4 = 1

Q75. Consider the standard result: Recursively enumerable vs recursive?

A. Recurs

Q76. Empty-string membership for RE language is?

A. Semi-d

Q77. Pumping length p for DFA with 4 states can be taken as?

A. 3

Q78. Consider the standard result: Kleene closure L* is regular if L is regular: proof uses?

A. PDA

Q79. Complement of CFL is?

A. Always

Q80. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Consta

Q81. Consider the standard result: A DFA is formally a 5-tuple (Q, ?, ?, q0, F) where ? is?

A. Q ? Q

Q82. L = {ww^R | w?{a,b}*} is?

A. Regula

Q83. For alphabet size 5, number of strings of length n is?

A. 5*n

Q84. Consider the standard result: ?-closure of a state includes?

A. Only st

Q85. Equivalence of two DFAs is decidable by?

A. Halting

Q86. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 17

Q87. Consider the standard result: Which language is NOT regular?

A. {a^n | n

Q88. Every finite language is?

A. Undeci

Q89. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 sta

Q90. Consider the standard result: Regular expressions, NFA, DFA are?

A. Equival

Q91. L = {a^n b^n c^n} is?

A. Contex

Q92. Pumping length p for DFA with 5 states can be taken as?

A. 5

Q93. Consider the standard result: Example of deterministic CFL but ambiguous grammar?

A. Halting

Q94. Intersection of CFL with regular is?

A. Undeci

Q95. Number of productions to convert CFG with 4 productions to CNF is bounded by?

A. Expone

Q96. Consider the standard result: Chomsky hierarchy from most restricted: Regular ? CFL ? CSL ? ?

A. Finite o

Q97. L = {a^p | p prime} is?

A. Star-fre

Q98. For alphabet size 5, number of strings of length n is?

A. 5+n

Q99. Consider the standard result: Pumping lemma for CFL uses?

A. xyz onl

Q100. Empty-string membership for RE language is?

A. Regula

Q101. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 32

Q102. Consider the standard result: Inherent ambiguity example?

A. (a+b)*

Q103. Complement of CFL is?

A. Not nec

Q104. If NFA has 4 states, worst-case DFA via subset construction has?

A. 4^2 sta

Q105. Consider the standard result: Turing machine adds over PDA?

A. Only st

Q106. L = {ww^R | w?{a,b}*} is?

A. Not CF

Q107. Pumping length p for DFA with 3 states can be taken as?

A. 3

Q108. Consider the standard result: An example non-computable / undecidable problem?

A. Determ

Q109. Equivalence of two DFAs is decidable by?

A. Halting

Q110. Number of productions to convert CFG with 3 productions to CNF is bounded by?

A. Polynor

Q111. Consider the standard result: Which CFL problem is undecidable?

A. Emptin

Q112. Every finite language is?

A. Not reg

Q113. For alphabet size 5, number of strings of length n is?

A. 5*n

Q114. Consider the standard result: A universal Turing machine?

A. Simulat

Q115. L = {a^n b^n c^n} is?

A. CFL

Q116. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 7

Q117. Consider the standard result: If L and its complement are both RE then L is?

A. Contex

Q118. Intersection of CFL with regular is?

A. Not CF

Q119. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5 states

Q120. Consider the standard result: Mealy vs Moore machines differ in?

A. Input al

Q121. L = {a^p | p prime} is?

A. Regula

Q122. Pumping length p for DFA with 4 states can be taken as?

A. 5

Q123. Consider the standard result: Every NFA can be converted to an equivalent DFA with at most?

A. 2^n sta

Q124. Empty-string membership for RE language is?

A. Always

Q125. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Polynor

Q126. Consider the standard result: Pumping lemma for regular languages states: for long enough string xyz with |y|>0, |xy|?p, then?

A. xy^iz ?

Q127. Complement of CFL is?

A. Always

Q128. For alphabet size 2, number of strings of length n is?

A. 2^n

Q129. Consider the standard result: Minimal DFA is unique up to isomorphism and found by?

A. Partitio

Q130. L = {ww^R | w?{a,b}*} is?

A. Not CF

Q131. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 65

Q132. Consider the standard result: A DPDA is strictly weaker than?

A. DFA

Q133. Equivalence of two DFAs is decidable by?

A. Halting

Q134. If NFA has 4 states, worst-case DFA via subset construction has?

A. 4^2 sta

Q135. Consider the standard result: Greibach Normal Form productions start with?

A. Variabl

Q136. Every finite language is?

A. CFL on

Q137. Pumping length p for DFA with 2 states can be taken as?

A. 3

Q138. Consider the standard result: Type-1 (context-sensitive) correspond to?

A. Linear

Q139. L = {a^n b^n c^n} is?

A. Finite

Q140. Number of productions to convert CFG with 4 productions to CNF is bounded by?

A. Undeci

Q141. Consider the standard result: Ambiguous grammar means?

A. No term

Q142. Intersection of CFL with regular is?

A. Undeci

Q143. For alphabet size 3, number of strings of length n is?

A. n^3

Q144. Consider the standard result: Pushdown automaton uses?

A. Tape o

Q145. L = {a^p | p prime} is?

A. Finite

Q146. The number of states in minimal DFA for language (0+1)*0(0+1){4} (k fixed suffix length) is at least?

A. 6

Q147. Consider the standard result: Halting problem is?

A. Decidal

Q148. Empty-string membership for RE language is?

A. Regula

Q149. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5 states

Q150. Consider the standard result: Decidable problems for DFA include?

A. Ambigu

Q151. Complement of CFL is?

A. Always

Q152. Pumping length p for DFA with 4 states can be taken as?

A. 3

Q153. Consider the standard result: Linear bounded automata decide?

A. Regula

Q154. L = {ww^R | w?{a,b}*} is?

A. Regula

Q155. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Expone

Q156. Consider the standard result: Complement of halting language is?

A. Regula

Q157. Equivalence of two DFAs is decidable by?

A. Minimiz

Q158. For alphabet size 2, number of strings of length n is?

A. 2*n

Q159. Consider the standard result: Arden's theorem solves?

A. CFGs

Q160. Every finite language is?

A. CFL on

Q161. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 7

Q162. Consider the standard result: An NFA transition function maps to?

A. Single

Q163. L = {a^n b^n c^n} is?

A. Contex

Q164. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 = 4

Q165. Consider the standard result: Language accepted by DFA is called?

A. Regula

Q166. Intersection of CFL with regular is?

A. Regula

Q167. Pumping length p for DFA with 4 states can be taken as?

A. 3

Q168. Consider the standard result: Myhill-Nerode theorem characterizes regularity via?

A. Infinite

Q169. L = {a^p | p prime} is?

A. Not reg

Q170. Number of productions to convert CFG with 3 productions to CNF is bounded by?

A. Undeci

Q171. Consider the standard result: Context-free grammars are recognized by?

A. Pushdo

Q172. Empty-string membership for RE language is?

A. Semi-d

Q173. For alphabet size 5, number of strings of length n is?

A. 5+n

Q174. Consider the standard result: Chomsky Normal Form productions are of type?

A. A?BC o

Q175. Complement of CFL is?

A. Always

Q176. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 8

Q177. Consider the standard result: Type-0 grammars correspond to?

A. DFA

Q178. L = {ww^R | w?{a,b}*} is?

A. Not CF

Q179. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 sta

Q180. Consider the standard result: Which language is CFL but not regular?

A. {a^n b^

Q181. Equivalence of two DFAs is decidable by?

A. Pumpir

Q182. Pumping length p for DFA with 3 states can be taken as?

A. 4

Q183. Consider the standard result: CYK algorithm decides membership for CFG in CNF in?

A. O(n^3)

Q184. Every finite language is?

A. Undeci

Q185. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Expone

Q186. Consider the standard result: Church-Turing thesis states?

A. NFA=D

Q187. L = {a^n b^n c^n} is?

A. Regula

Q188. For alphabet size 2, number of strings of length n is?

A. 2^n

Q189. Consider the standard result: Rice's theorem says?

A. P=NP

Q190. Intersection of CFL with regular is?

A. Undeci

Q191. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 5

Q192. Consider the standard result: Post Correspondence Problem (PCP) is?

A. Finite

Q193. L = {a^p | p prime} is?

A. Star-fre

Q194. If NFA has 3 states, worst-case DFA via subset construction has?

A. 3^2 sta

Q195. Consider the standard result: Recursively enumerable vs recursive?

A. Recurs

Q196. Empty-string membership for RE language is?

A. Semi-d

Q197. Pumping length p for DFA with 4 states can be taken as?

A. 2k

Q198. Consider the standard result: Kleene closure L* is regular if L is regular: proof uses?

A. Pumpir

Q199. Complement of CFL is?

A. Always

Q200. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Expone

ANSWER KEY

1-D, 2-D, 3-C, 4-D, 5-D, 6-B, 7-C, 8-A, 9-D, 10-C

11-C, 12-B